PHYSIOLOGY

Acid–Base Balance by the Kidneys

SGL 2
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GUE BLOCK

Physiology GUE Block · MEDucation by Mattin Majid

BEFORE YOU READ — ANSWER FROM MEMORY

1. What is the difference between reclaiming filtered bicarbonate and generating new bicarbonate — and which one actually adds HCO_3^- to the blood?
2. Write the Henderson–Hasselbalch equation and name each term.
3. What is the net acid excretion (NAE) formula, and which urinary buffer does most of the adaptive work during acidosis?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Who removes what — CO_2 vs nonvolatile acid

Metabolism	Proteins/carbs/fats are metabolized for ATP; the removed hydrogens and carbons recombine to form **CO_2**
CO_2 handling	$CO_2 + H_2O \rightleftharpoons H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$ (carbonic anhydrase). **Lungs** remove CO_2 by exhalation
Nonvolatile acid	**Kidneys** remove bicarbonate and H^+ — net nonvolatile acid — via NH_4^+ + titratable acid excretion and HCO_3^- conservation/replacement

Term	Definition
Acid	A molecule that can **release H^+** in solution (e.g. $HCl \rightarrow H^+ + Cl^-$; $H_2CO_3 \rightarrow H^+ + HCO_3^-$)
Base	An ion/molecule that can **accept H^+** (e.g. HCO_3^-, HPO_4^{2-}, plasma proteins like hemoglobin)

2 pH & the Henderson–Hasselbalch equation

pH	$[H^+]$, nmol/L
7.30	≈50
7.40	≈40
7.50	≈32

$pH = -\log_{10}[H^+]$. A fall of **one pH unit = a tenfold rise in $[H^+]$**. Normal arterial pH: **7.35–7.45**. **Acidemia <7.35; Alkalemia >7.45.** Acidosis/alkalosis describe the underlying *process*, not necessarily the measured pH.

HENDERSON–HASSELBALCH

$pH = 6.1 + \log_{10}\left([HCO_3^-] \div (0.03 \times PCO_2) \right)$ — 6.1 is the apparent pK near 37°C; $[HCO_3^-]$ in mmol/L; PCO_2 in mmHg; 0.03 is CO_2 solubility ($mmol \cdot L^{-1} \cdot mmHg^{-1}$).

Example: $24 \div (0.03 \times 40) = 20 \rightarrow pH = 6.1 + \log_{10}20 \approx$ **7.40**

3 Three defenses against pH change

Defense	Time	Action
Chemical buffers	Seconds	Bind or release H^+ (bicarbonate, proteins, phosphate)
Lungs	Minutes	Adjust CO_2 elimination
Kidneys	Hours to days	**Reclaim HCO_3^-** and adjust **net acid excretion**

These are overlapping, ongoing responses — not durations after which a defense "stops."

4 Renal bicarbonate handling — the kidney's 3 jobs

FILTERED LOAD

180 L/day × 24 mmol/L ≈ 4,320 mmol of HCO_3^- filtered per day (illustrative GFR × plasma concentration)

1	**Reclaim** filtered bicarbonate — protects the existing HCO_3^- pool
2	**Replace** bicarbonate consumed by buffering nonvolatile acid
3	**Excrete acid** bound to urinary buffers (titratable acid, NH_4^+)

Site	Fraction of filtered HCO_3^- reclaimed	Mechanism
Proximal tubule	≈80–90%	NHE3 (apical Na^+/H^+ exchange); ammonia production begins here
Thick ascending limb	≈10–15%	NH_4^+ recycling
Late distal nephron / collecting duct	Remainder	Final H^+ secretion

Proximal reclamation — the mechanism

①	Filtered HCO_3^- + secreted H^+ (via apical **NHE3**) → H_2CO_3 in the lumen
②	Luminal carbonic anhydrase (CA IV) → CO_2 + H_2O, which diffuses into the cell
③	Intracellular CA II: CO_2 + H_2O → H^+ + HCO_3^- (H^+ recycled back out via NHE3)
④	HCO_3^- exits basolaterally via **NBCe1** (with Na^+) toward the blood

Net effect: a filtered HCO_3^- equivalent returns to blood; **H^+ is recycled**, not consumed. Na^+/K^+-ATPase maintains the Na^+ gradient that powers this.

5 Reclamation vs new bicarbonate — THE key distinction

	Reclamation	New bicarbonate generation
Reaction	H^+ + filtered HCO_3^-	H^+ + nonbicarbonate buffer (phosphate or ammonia)
What happens	Filtered HCO_3^- equivalent returns to blood	H^+ leaves in urine; a **new** HCO_3^- enters blood
Net new HCO_3^-	**Zero**	**One equivalent**

For ammonium specifically: urinary **excretion** must follow renal **production** to yield a net systemic gain in bicarbonate.

6 The alpha-intercalated cell (distal H^+ secretion)

Apical: **H^+-ATPase** and **H^+/K^+-ATPase** (ATP-powered) secrete H^+ into the lumen. Intracellular CA II forms H^+ + HCO_3^- from CO_2+H_2O. Basolateral **AE1** (anion exchanger) sends the generated HCO_3^- to blood in exchange for Cl^-.

In **alkalosis**, **beta-intercalated cells** reverse the polarity: apical **pendrin** secretes HCO_3^- into the lumen (exchanged for Cl^- into the cell) — this is how the kidney excretes excess bicarbonate.

7 Urinary buffers — phosphate vs ammonium

Feature	Phosphate / titratable acid	Ammonium
Origin	Filtered phosphate buffer	Proximal glutamine metabolism
Urinary product	$H_2PO_4^-$	Mostly NH_4^+
Response to acidosis	**Limited** by available filtered buffer	**Large adaptive increase** in production
New bicarbonate yield	1 equivalent per net H^+ excreted	1 equivalent per NH_4^+ excreted

HPO_4^{2-} + H^+ ⇌ $H_2PO_4^-$ (titratable acid). **Urine pH measures H^+ activity, not the total amount of acid excreted** — a low urine pH alone cannot carry the full daily acid load; phosphate's supply is limited, so **ammonium does most of the adaptive work** during acidosis (via increased glutamine metabolism → NH_4^+, with NH_4^+ recycled at the thick ascending limb and transferred across the collecting duct, partly via Rhcg).

NET ACID EXCRETION (NAE)

NAE = NH_4^+ + Titratable acid − HCO_3^- (all terms are urinary excretion rates)

Example: 40 + 20 − 5 = **55 mEq/day**

8 Renal response to acidosis vs alkalosis

Variable	Acidosis	Alkalosis (if renal response intact)
HCO_3^- handling	Conserve filtered HCO_3^-	Permit HCO_3^- excretion
H^+ secretion / NAE	Usually increases	Usually decreases
NH_4^+ excretion	Increases with adaptation	Usually decreases

Volume depletion, low chloride, high potassium, and reduced nephron number can all blunt or prevent the "expected" renal response — e.g. angiotensin II/low volume maintains alkalosis by boosting proximal HCO_3^- reclamation; high plasma K^+ suppresses ammoniagenesis (type 4 RTA); fewer functioning nephrons (CKD) lowers total acid-excretory capacity.

9 Interpreting an acid–base disturbance

1	Classify pH: acidemia, reference range, or alkalemia
2	Identify the **primary** HCO_3^- or PCO_2 change
3	Calculate the **expected compensation**
4	Check electrolytes, time course, and clinical context

Anion gap = $Na^+ - (Cl^- + HCO_3^-)$

Primary process	Expected compensation (approximate)
Metabolic acidosis	$PCO_2 = 1.5 \times [HCO_3^-] + 8 \pm 2$ (Winter's rule)
Metabolic alkalosis	$PCO_2 \approx 40 + 0.7 \times ([HCO_3^-] - 24)$
Respiratory acidosis	HCO_3^- rises ≈1 (acute) or 3–4 (chronic) per 10↑ PCO_2
Respiratory alkalosis	HCO_3^- falls ≈2 (acute) or 4–5 (chronic) per 10↓ PCO_2

PCO_2 in mmHg; HCO_3^- in mmol/L. Baseline 40/24. Chronic adaptation takes days. These rules **screen for an added process** — they are not exact laws or treatment targets.

P Must-remember pearls

- **Reclaiming** filtered HCO_3^- gives **zero net new bicarbonate**; buffering H^+ with phosphate or ammonia gives **one new HCO_3^- equivalent** to blood — the single most testable distinction in this lecture.
- **Henderson–Hasselbalch: $pH = 6.1 + \log_{10}([HCO_3^-] \div (0.03 \times PCO_2))$** — normal 24/(0.03×40) = 20 → pH ≈7.40.
- **NAE = NH_4^+ + Titratable acid − HCO_3^-.** Phosphate (titratable acid) is **limited** by filtered supply; **ammonium production rises dramatically** in acidosis via proximal glutamine metabolism.
- The **thick ascending limb** reclaims ≈10–15% of filtered HCO_3^- and recycles NH_4^+; the **proximal tubule** reclaims the bulk (≈80–90%) via NHE3.
- Normal arterial pH **7.35–7.45**; acidemia <7.35, alkalemia >7.45. A **normal pH can still conceal two opposing disturbances** — always check the anion gap and expected compensation.

Q Test yourself

1. A filtered HCO_3^- ion combines with a secreted H^+ in the tubular lumen, is converted to CO_2 and water, and the equivalent HCO_3^- re-enters the blood via NBCe1. What is the net new bicarbonate generated by this process?

A. One new HCO_3^- equivalent
B. Zero — this is reclamation, not new bicarbonate generation
C. Two new HCO_3^- equivalents
D. It depends on urine pH
E. Negative — bicarbonate is lost

2. Which apical transporter in the alpha-intercalated cell directly uses ATP to secrete H^+?

A. NHE3
B. AE1
C. H^+-ATPase
D. NBCe1
E. NKCC2

3. A patient's urine shows NH_4^+ excretion of 40 mEq/day, titratable acid of 20 mEq/day, and HCO_3^- excretion of 5 mEq/day. What is the net acid excretion?

A. 15 mEq/day
B. 55 mEq/day
C. 60 mEq/day
D. 65 mEq/day
E. 5 mEq/day

4. A patient has chronic metabolic acidosis with plasma HCO_3^- of 14 mmol/L and PCO_2 of 40 mmHg. Using Winter's rule ($PCO_2 = 1.5 \times HCO_3^- + 8 \pm 2$), what is the interpretation?

A. Appropriate respiratory compensation
B. An additional respiratory acidosis is present (PCO_2 higher than expected)
C. An additional respiratory alkalosis is present
D. This represents isolated metabolic alkalosis
E. No compensation has occurred at all

5. Why can urine pH alone not be used to estimate the total amount of acid excreted per day?

A. Urine pH measures H^+ activity, not the total titratable + ammonium acid load
B. Urine pH is always 7.4 regardless of acid load
C. Ammonium does not affect urine pH
D. Phosphate buffering has unlimited capacity
E. Urine pH only reflects bicarbonate excretion

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 — B — this describes reclamation: the filtered HCO_3^- equivalent returns to blood, but no new bicarbonate is created.

2 — C — the H^+-ATPase (and H^+/K^+-ATPase) on the apical membrane are ATP-powered; NHE3 and NBCe1 are proximal tubule transporters, AE1 is basolateral.

3 — B — NAE = NH_4^+ + titratable acid − HCO_3^- = 40 + 20 − 5 = 55 mEq/day.

4 — B — expected $PCO_2 = 1.5 \times 14 + 8 = 29 \pm 2$ mmHg (27–31). An observed PCO_2 of 40 mmHg is higher than expected, indicating a superimposed respiratory acidosis.

5 — A — urine pH reflects free H^+ activity only; most excreted acid is bound to titratable buffers and ammonium, which pH alone does not quantify.

SPACED REVIEW

☐ Day 1 | ☐ Day 3 | ☐ Day 7 | ☐ Day 21 | ☐ Pre-exam